

**This assignment will not be graded and is only for practice.**

### Level 1

1) Consider the following expression:  $v_1 = (1, 0)$ ,  $v_2 = (0, 0)$ ,  $v_3 = (0, 1)$ ,  $v_5 = 1$ ,  $v_6 = 5$ ,  $v_7 = (1, 0, 1)$ ,  $v_8 = (0, 0, 1)$ ,  $v_9 = (0, 0, 0)$ ,  $v_{10} = (0, 1, 0)$ . Choose the set of correct options. **1 point**

- ☐  $v_1, v_7$  represent vectors in  $\mathbb{R}^2$ .
- ☐  $v_2, v_5$  represent vectors in  $\mathbb{R}^1$ .
- ☐  $v_2, v_5, v_9$  represent vectors in  $\mathbb{R}^3$ .
- ☐  $v_1, v_2, v_3$  represent vectors in  $\mathbb{R}^2$ .
- ☐  $v_5, v_6$  represent vectors in  $\mathbb{R}^1$ .
- ☐  $v_1, v_6$  represent vectors in  $\mathbb{R}^1$ .
- ☐  $v_8, v_9, v_{10}$  represent vectors in  $\mathbb{R}^3$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$v_1, v_2, v_3$  represent vectors in  $\mathbb{R}^2$ .  
 $v_5, v_6$  represent vectors in  $\mathbb{R}^1$ .  
 $v_8, v_9, v_{10}$  represent vectors in  $\mathbb{R}^3$ .

2) Let  $v, v_1, v_2$  represent vectors in a vector space  $V$  over  $\mathbb{R}$ . Which of the following expressions make sense and represent vectors in  $V$ ? **1 point**

- ☐  $cv \in V$ , where  $c \in \mathbb{R}$ .
- ☐  $\frac{v}{2}$ .
- ☐  $v^2$ .
- ☐  $v_1 - v_2$ .
- ☐  $v_1^2 - v_2^2$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$cv \in V$ , where  $c \in \mathbb{R}$ .

$\frac{v}{2}$ .

$v_1 - v_2$ .

3) Consider the set  $V = \{(z - x, -y) \mid x, y, z \in \mathbb{R}\} \subseteq \mathbb{R}^2$  with the usual addition and scalar multiplication as in  $\mathbb{R}^2$  and associated statements given below. Choose the correct options. **1 point**

- ☐  $V$  is closed under addition i.e  $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$ .
- ☐  $V$  is closed under scalar multiplication i.e  $(c \in \mathbb{R}, v \in V \implies cv \in V)$ .
- ☐  $V$  has a zero element with respect to addition. i.e., there exists some element  $0$  such that  $v + 0 = v$ , for all  $v \in V$ .
- ☐  $V$  is not a vector space.
- ☐  $(a + b)v = av + bv$  where  $a, b \in \mathbb{R}$  and  $v \in V$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V$  is closed under addition i.e  $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$ .

$V$  is closed under scalar multiplication i.e  $(c \in \mathbb{R}, v \in V \implies cv \in V)$ .

$V$  has a zero element with respect to addition. i.e., there exists some element  $0$  such that  $v + 0 = v$ , for all  $v \in V$ .

$(a + b)v = av + bv$  where  $a, b \in \mathbb{R}$  and  $v \in V$ .

4) Consider a set  $V = \{(1, x) \mid x \in \mathbb{R}\} \subseteq \mathbb{R}^2$  with the usual addition and scalar multiplication as in  $\mathbb{R}^2$  and associated statements given below. Choose the correct options. **1 point**

- ☐  $V$  is closed under addition.
- ☐  $V$  has no zero element with respect to addition. i.e., there does not exists any element  $0$  such that  $v + 0 = v$ , for all  $v \in V$ .
- ☐  $(a + b)v = av + bv$  where  $a, b \in \mathbb{R}$  and  $v \in V$ .

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V$  has no zero element with respect to addition. i.e., there does not exist any element  $0$  such that  $v + 0 = v$ , for all  $v \in V$ .

$(a + b)v = av + bv$  where  $a, b \in \mathbb{R}$  and  $v \in V$ .

5) Which of the following sets with the given addition and scalar multiplication (scalars are real **1 point** numbers in every case) form vector spaces?

$$V_1 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$$

$$\text{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + 1, z_2 + 1);$$

$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_1$$

$$\text{Scalar multiplication: } c(x, y, z) = (x, y, cz); (x, y, z) \in V_1, c \in \mathbb{R}$$

$$V_2 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$$

$$\text{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1, y_1, z_1);$$

$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_2$$

$$\text{Scalar multiplication: } c(x, y, z) = (x, cy, z); (x, y, z) \in V_2, c \in \mathbb{R}$$

$$V_3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$$

$$\text{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, 0, 0);$$

$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_3$$

$$\text{Scalar multiplication: } c(x, y, z) = (cx, y, z); (x, y, z) \in V_3, c \in \mathbb{R}$$

Choose the correct options.

☐  $V_3$  is not vector space.

☐  $V_2$  is vector spaces.

☐ All are vector spaces.

☐  $V_1$  is not vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V_3$  is not vector space.

$V_1$  is not vector space.

6) If  $(a, b, c)$  is the inverse element of  $(1, 2, 5) \in \mathbb{R}^3$  of the vector space  $\mathbb{R}^3$ , then find the value of  $a - b + c$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -4

**1 point**

## Level 2

7) Consider the following sets:

**1 point**

$V_1 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a symmetric matrix, i.e., } A = A^T\}$

$V_2 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a scalar matrix}\}$

$V_3 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a diagonal matrix}\}$

$V_4 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a upper triangular matrix}\}$

$V_5 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a lower triangular matrix}\}$

Choose the set of correct options.

- ☐ Only  $V_1$  is a subspace of  $M_{2 \times 2}(\mathbb{R})$ .
- ☐ Only  $V_4$  is a subspace of  $M_{2 \times 2}(\mathbb{R})$ .
- ☐ Both  $V_2$  and  $V_3$  are subspaces of  $M_{2 \times 2}(\mathbb{R})$
- ☐ All are subspaces of  $M_{2 \times 2}(\mathbb{R})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both  $V_2$  and  $V_3$  are subspaces of  $M_{2 \times 2}(\mathbb{R})$

All are subspaces of  $M_{2 \times 2}(\mathbb{R})$ 

8) Choose the set of correct options.

**1 point**

- ☐ A vector in  $\mathbb{R}^3$  can be thought of as matrix of order  $1 \times 3$  or  $3 \times 1$ .
- ☐ Intersection of any two subspaces of a vector space  $V$  is not a subspace.
- ☐ Intersection of any two subsets of a vector space  $V$  is a subspace.
- ☐ The empty subset of a vector space  $V$  is a subspace of  $V$ .
- ☐  $V = \{(x, 0, 0) \mid x \in \mathbb{R}\}$  is a subspace of  $\mathbb{R}^3$ .

No, the answer is incorrect.

Score: 0

## Accepted Answers:

A vector in  $\mathbb{R}^3$  can be thought of as matrix of order  $1 \times 3$  or  $3 \times 1$ . $V = \{(x, 0, 0) \mid x \in \mathbb{R}\}$  is a subspace of  $\mathbb{R}^3$ .9) Consider a set  $V = \{(x, y) \mid x, y \in \mathbb{R}\} \subseteq \mathbb{R}^2$  with the usual addition as in  $\mathbb{R}^2$  and scalar multiplication is defined as**1 point**

$$c(x, y) = \begin{cases} (0, 0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}) & c \neq 0 \end{cases} \quad (x, y) \in V, c \in \mathbb{R}$$

Consider the statements given below.

**P:**  $V$  is closed under addition.**Q:**  $V$  has zero element with respect to addition. i.e., there exists some element  $0$  such that  $v + 0 = v$ , for all  $v \in V$ .**R:**  $1 \cdot v = v$  where  $1 \in \mathbb{R}$  and  $v \in V$ .**S:**  $a(v_1 + v_2) = av_1 + av_2$  where  $v_1, v_2 \in V$  and  $a \in \mathbb{R}$ .**T:**  $(a + b)v = av + bv$  where  $a, b \in \mathbb{R}$  and  $v \in V$ .

Choose the set of correct options.

- ☐ Only P is true.
- ☐ Only Q is true.
- ☐ Both P and Q are true.

- ☐ P,Q and S are true.
- ☐ Both R and T are not true.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both P and Q are true.  
 P,Q and S are true.  
 Both R and T are not true.

10) Consider the following statements

**Statement P:**  $W = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A^2 = A\}$ , a subset of  $M_{2 \times 2}(\mathbb{R})$  with usual addition and scalar multiplication as in  $M_{2 \times 2}(\mathbb{R})$ , is a subspace of  $M_{2 \times 2}(\mathbb{R})$ .

**Statement Q:**  $W = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } \det(A) = 0\}$ , a subset of  $M_{2 \times 2}(\mathbb{R})$  with usual addition and scalar multiplication as in  $M_{2 \times 2}(\mathbb{R})$ , is a subspace of  $M_{2 \times 2}(\mathbb{R})$ .

**Statement R:** Consider a system of linear equations  $Ax^T = b$ , where  $A$  is invertible square matrix of order 2,  $x = (x_1, x_2)$  and  $b = 0$ . Let  $W = \{x = (x_1, x_2) \mid x \in \mathbb{R}^2 \text{ and } Ax^T = b\}$  is a subset of  $\mathbb{R}^2$  with usual addition and scalar multiplication as in  $\mathbb{R}^2$ , is a subspace of  $\mathbb{R}^2$ .

**Statement S:** Consider a system of linear equations  $Ax^T = 0$ , where  $A$  is invertible square matrix of order 2 and  $x = (x_1, x_2)$ . Let  $W = \{x = (x_1, x_2) \mid x \in \mathbb{R}^2 \text{ and } Ax^T = 0\}$  is a subset of  $\mathbb{R}^2$  with usual addition and scalar multiplication as in  $\mathbb{R}^2$ , is a subspace of  $\mathbb{R}^2$ .

Find the number of correct statements

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

**1 point**

Your score is: 0/10